ASRC Fuel Tank Robot Daily Report

Backbone M module: entry, stability, tendon force, 3D tank and horizontal flexibility, 9 November 2026

Yesterday's review showed that a central 1.8 mm NiTi rod in each M module lets U4 plus M2 straighten within the 5% strain diagnostic and hold the 500 g head near 20 degrees. Today tested whether that design also enters the tank, stays stable in 3D, can be steered with small and smooth tendon forces, and fits inside the full 3D tank.

Summary of today's work

- **Tank entry, low strain.** The straight rod shape feeds through the bottom opening at 15, 20 and 25 degrees with 44, 64 and 17 mm minimum clearance at about 0.1% strain. On 6 November every low-strain shape hit the tank.
- **3D stability needs pretension.** With only the up ropes pulled, the straight front is laterally unstable in 3D; the 2D model could not show this. A 1 N pretension on every rope fixes it, and straightening and entry still pass.
- **Minimum tendon force.** Holding the head at 20 degrees while sweeping yaw from 0 to 175 degrees in 5 degree steps needs only about 7 to 15 N peak rope force (anchors [4,5,6]), against 46 to 98 N for yesterday's allocations. With smoothing, no rope changes by more than about 5 N per step.
- **Full 3D tank.** All 362 tank solids were meshed from the STEP file. Entry clearance is 47 / 65 / 19 mm in full 3D. At 20 degrees with the clamp 0 to 30 mm past the opening, the whole yaw sweep clears the tank by at least 12.6 mm on both sides; 25 degrees hits the roof.
- **Horizontal flexibility.** Reeling in the tip's right rope turns the head about 1.25 degrees per mm, smoothly and stably. Alone it keeps the head at 20 plus or minus 5 degrees only to about 20 degrees of yaw. Tension-only control snaps the straight front (258 mm jump), so rope-length control is required.

Check (U4 plus M2, 1.8 mm NiTi rod)	Result	Status
Straightening at or below 5% strain, 15 to 25 deg	1.2 to 2.0 mm offset, about 0.1% strain, with 1 N pretension	Pass
Tank entry (2D sections and full 3D)	44 to 65 mm at 15 and 20 deg; 17 to 19 mm at 25 deg	Pass
3D lateral stability	Unstable with up ropes only; stable with 1 N on every rope	Pass with rule
Held-elevation yaw sweep, 20 deg	0 to 175 deg, 7 to 15 N peak, smooth 5 deg steps	Pass
Yaw sweep inside full 3D tank	20 deg, clamp s = 0 to 30 mm: at least 12.6 mm	Pass
25 deg installation, yaw sweep	Hits the tank roof at every clamp depth	Fail
Side pull of one rope, winch control	About 1.25 deg/mm; holds 20 deg elevation only to about 20 deg yaw	Needs coordination
Tension-only control of the straight front	Snap-through at +0.5 N	Do not use

Decisions from today

Install at 20 degrees. Steer with the clamp 0 to 30 mm past the opening. Keep at least 1 N on every rope. Drive the ropes by length (winch position) control, not tension control, and coordinate the up and side ropes to hold the working elevation.

Tank entry with the straight low-strain shape

The 6 November front-only entry screen was reused unchanged: a fixed loaded shape slides along the installation axis through the bottom opening, starting fully outside the tank, and must stay at least 5 mm from the saved tank sections. The control case reproduces the 6 November value exactly (62.06 mm). Passing shapes were re-checked at 16 cells per module.

U4 plus M2	Strain	15 deg	20 deg	25 deg
Current M (no rod)	4.96%	hits tank	hits tank	hits tank
Rod, E 40 GPa	0.10%	44.3 mm	64.0 mm	16.9 mm
Rod, E 75 GPa	0.09%	44.5 mm	64.4 mm	16.2 mm

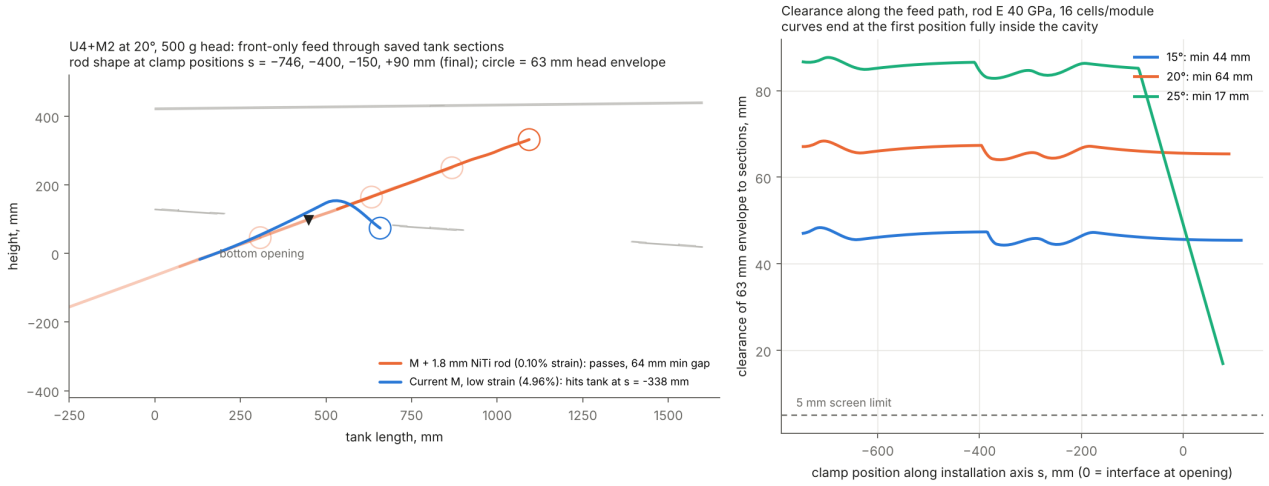


Figure 1 Left: the rod shape at four clamp positions through the saved tank sections at 20 degrees; the current-M low-strain shape hits the tank. Right: clearance along the feed path for 15, 20 and 25 degrees (16 cells per module).

3D stability needs pretension on every rope

The straightened states tension only the up ropes. They pass the planar stability check, but in the 3D model they are laterally unstable (minimum fixed-winch eigenvalue minus 0.20 to minus 0.50), because the 2D model cannot see sideways buckling. A 1 N pretension on all four ropes of every set stabilises every case. With it, the straightening was re-optimised:

Rod	Pitch	Offset / tangent / strain	Top-rope tensions N	3D stable	Entry gap
E 40 GPa	15 deg	1.86 mm / 4.2 deg / 0.11%	95.1, 21.8, 8.7	Yes	44.6 mm
E 40 GPa	20 deg	1.82 mm / 4.0 deg / 0.11%	92.7, 21.2, 8.5	Yes	64.2 mm
E 40 GPa	25 deg	2.00 mm / 3.5 deg / 0.10%	89.5, 20.5, 8.2	Yes	17.2 mm
E 75 GPa	15 deg	1.28 mm / 3.3 deg / 0.10%	87.7, 21.9, 8.7	Yes	44.7 mm
E 75 GPa	20 deg	1.25 mm / 3.1 deg / 0.10%	85.4, 21.3, 8.5	Yes	64.6 mm
E 75 GPa	25 deg	1.21 mm / 3.0 deg / 0.10%	82.4, 20.6, 8.2	Yes	16.1 mm

Conclusion: the 8 November straightening and today's entry results stand, provided every rope keeps at least 1 N. This becomes a control rule.

Minimum tendon force for held-elevation yaw

Yaw was stepped from 0 to 180 degrees by continuation, each step seeded from the previous one. At each step the optimiser minimised the largest single rope tension subject to equilibrium, the target yaw, head elevation held at the base pitch, the geometry limits, strain within 4.75% and 1 to 250 N per rope. States were solved at 4 cells per module, then re-solved at 8 cells and accepted only if stable and within 1 degree of the target. A small penalty on tension changes between steps selects a smooth allocation among the many with nearly the same peak.

Run (elevation held exactly, 5 deg steps)	Passing states	Peak rope N	Largest change per step
E 40 GPa, 20 deg, anchors [4,5,6], smooth	36/37 (0 to 175 deg)	7.1 to 14.8	5.1 N
E 40 GPa, 20 deg, anchors [4,5,6]	37/37 (0 to 180 deg)	7.0 to 14.8	13.8 N
E 40 GPa, 20 deg, anchors [4,6]	36/37 (0 to 175 deg)	13.3 to 22.0	20.9 N
E 75 GPa, 20 deg, anchors [4,5,6], smooth	37/37 (0 to 180 deg)	9.8 to 14.6	10.0 N
E 40 GPa, 25 deg, anchors [4,5,6]	37/37 (0 to 180 deg)	6.7 to 13.3	12.3 N

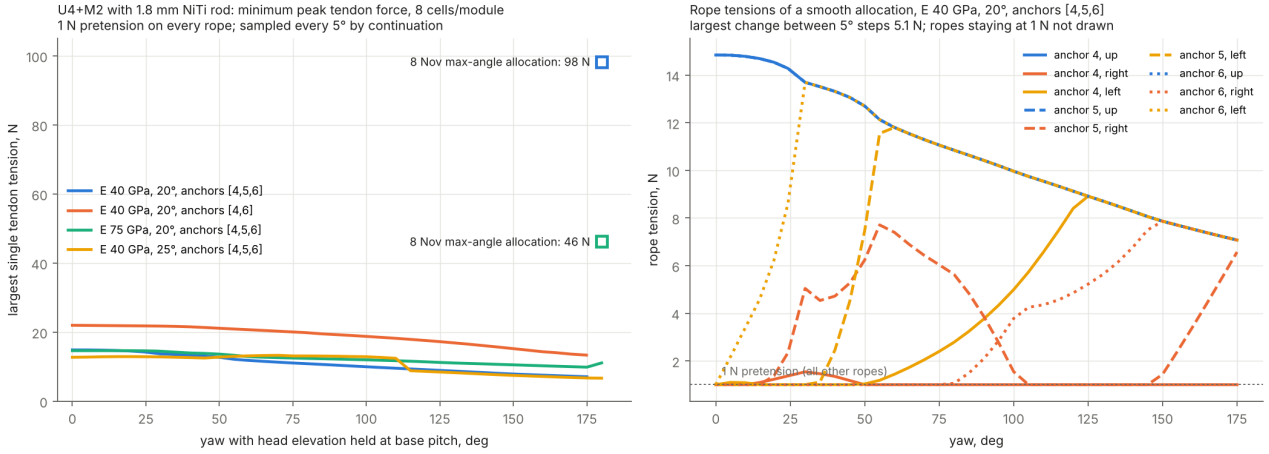


Figure 2 Left: minimum peak rope tension along the held-elevation yaw sweep, against the 8 November maximum-angle allocations. Right: rope tensions of the smooth allocation.

- Peak force drops from 46 to 98 N to about 7 to 15 N with anchors [4,5,6]. Strain stays at or below 0.09%, and every state is stable.
- Consecutive 5 degree states move the head at most 33 to 56 mm, consistent with a smooth sweep. This is strong evidence of a continuous quasi-static path from 0 to 175 degrees, but the states between samples were not checked.
- Posture: 30 to 90 degrees of yaw uses sideways bending of M with 30 to 55 degrees of twist; near 180 degrees M curls up and back over itself, so the head looks back toward the base. Up to about 60 degrees the head stays within about 20 mm of its starting height.

Full 3D tank clearance

All 362 solids of the tank model in tank-v2.step (external frame excluded) were triangulated: 461,524 triangles at 0.5 mm deflection. Clearance is the exact distance from the front centreline to that surface minus the 31.5 mm envelope radius. Points on the five saved 2D sections lie within 0.23 mm of the 3D mesh.

Entry path, straight shape, 1 N pretension	15 deg	20 deg	25 deg
2D sections (five planes)	44.6 mm	64.2 mm	17.2 mm
Full 3D tank	46.8 mm	64.6 mm	19.3 mm

The full-width 3D result confirms the 2D screen and closes the limitation that five planes did not cover the full 63 mm width.

Held-elevation yaw sweep placed in the 3D tank, by clamp depth s past the opening

20 deg, head held at 20 deg	s = 0 mm	s = 30 mm	s = 60 mm	s = 90 mm
E 40 GPa, right	all clear, min 22.4 mm	all clear, min 12.6 mm	33/36 clear, min 2.8 mm	26/36 clear, min -6.9 mm
E 40 GPa, left	all clear, min 30.9 mm	all clear, min 20.9 mm	all clear, min 11.0 mm	33/36 clear, min 1.6 mm
E 75 GPa, right	all clear, min 23.5 mm	all clear, min 13.6 mm	35/37 clear, min 3.7 mm	25/37 clear, min -6.2 mm
E 75 GPa, left	all clear, min 30.1 mm	all clear, min 20.1 mm	all clear, min 10.2 mm	34/37 clear, min 0.8 mm

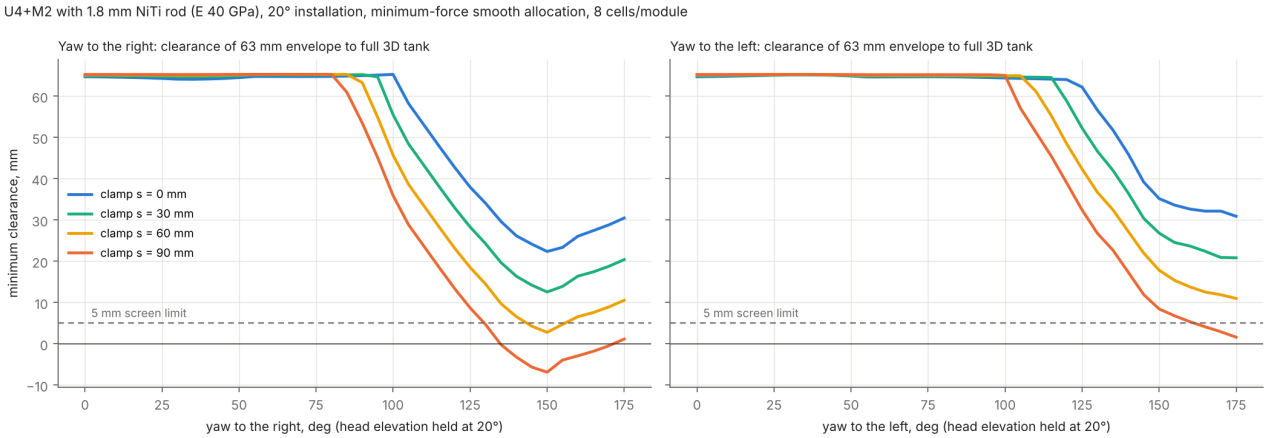


Figure 3 Clearance to the full 3D tank along the held-elevation yaw sweep, E 40 GPa, by clamp depth.

- At 20 degrees with the clamp 0 to 30 mm past the opening, the whole sweep clears the tank by at least 12.6 mm on both sides. Clearance stays near 65 mm up to about 85 degrees of yaw and falls only in the up-and-over posture, always against the tank roof.
- Deeper clamp positions raise the base and bring the curled M into the roof. At 25 degrees the sweep fails at every depth tested (s = 0 to 300 mm).

Horizontal flexibility: yaw vs side-rope pull

Starting from the straight held front at 20 degrees, only the right rope was pulled, at anchor 6 (tip), 5, 4 or all three, in small steps from the previous state at 8 cells per module. Two control modes were compared. Tension control prescribes rope forces. Winch control prescribes rope lengths: each rope is an elastic PA6 line (EA 10.4 kN) with 1 m of free length, reeled in 0.25 mm per step.

Tension-only control is unsafe for the straight pose
At constant tension the straight front is laterally unstable (eigenvalue minus 0.51 at E 40 GPa). Adding 0.5 N to the tip's right rope makes the head snap 258 mm to 124 degrees of yaw. Under winch control the same pose is stable (eigenvalue plus 0.73). Use rope-length control or fast feedback.

Pulled rope (winch)	Initial gain	Yaw with head at 20 plus or minus 5 deg	Largest yaw (elevation, rope N)	Largest head jump
Anchor 6, E 40 GPa	1.27 deg/mm	18.6 deg (14.25 mm, 37 N)	58.5 deg (-41 deg, 27 N)	1.0 mm
Anchor 6, E 75 GPa	1.23 deg/mm	20.2 deg (15.75 mm, 43 N)	76.1 deg (-49 deg, 39 N)	1.0 mm
All three, E 40 GPa	1.36 deg/mm	16.9 deg (13.5 mm, 65 N)	27.3 deg (-17 deg, 142 N)	4.3 mm
Anchor 5, E 40 GPa	0.12 deg/mm	7.9 deg (16 mm, 37 N)	33.8 deg (-34 deg, 13 N)	1.6 mm
Anchor 4, E 40 GPa	-0.02 deg/mm	3.1 deg (33 mm, 154 N)	8.8 deg (+2 deg, 288 N)	1.3 mm

U4+M2 with 1.8 mm NiTi rod, 20°, start: straight held front (1 N pretension); winch (rope-length) control, 8 cells/module, free space

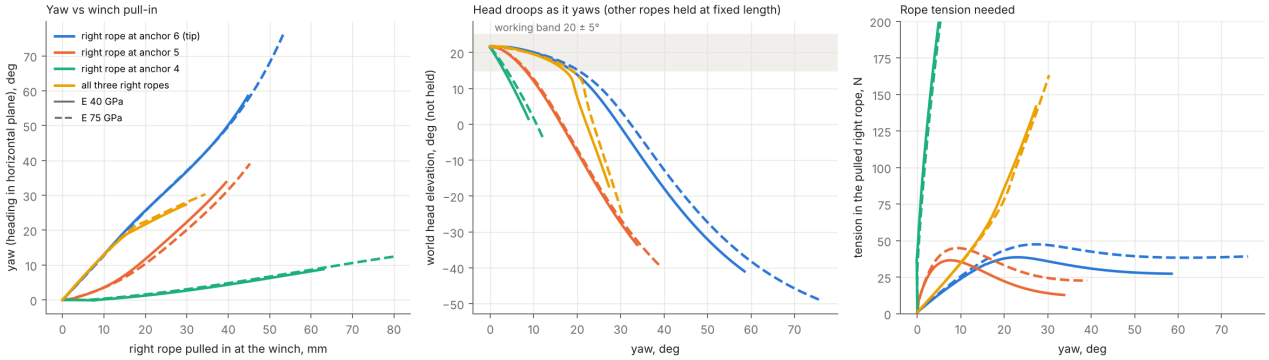


Figure 4 Yaw against winch pull-in, head elevation against yaw, and tension in the pulled rope.

- The tip rope is the effective yaw actuator: about 1.27 deg per mm of pull-in, roughly 2 N per degree at small angles, smooth and stable.
- Anchors 4 and 5 hardly turn the head: the U section barely bends sideways, so the force compresses the front instead (up to 364 N for 12 degrees at anchor 4).
- With the other ropes locked, the head droops as it yaws. It stays at 20 plus or minus 5 degrees only to about 19 to 20 degrees. Further yaw at the working elevation needs the up ropes reeled in together with the side rope, as the minimum-force sweep does.

Limitations

- All rod properties are calculated, not measured. Constant linear EA and EI; no friction, NiTi transformation, hysteresis or fatigue.
- The sensor head is a 500 g point mass with a provisional 63 mm envelope. Cables and ropes outside the front are not modelled.
- Steering states are free-space equilibria placed in the tank, not tank-aware optimised postures. The 30 to 55 degrees of twist used at mid yaw is unverified in hardware.
- States between the 5 degree samples, and real winch dynamics, were not checked.
- The 5% strain diagnostic, 250 N rope bound and 180 degree search cap are numerical limits, not hardware ratings.

Next steps

Priority	Task	Purpose
1	Build and test the backbone M module (test plan)	Measure EA, EI, buckling at the guide spacing and rod joint slip; everything above depends on these
2	Coordinated winch-control test (planned for tomorrow)	Drive the rope lengths through the 5 degree minimum-force poses with fine steps; check the head moves smoothly from 0 to 175 degrees at 20 degrees elevation
3	Tank-aware steering	Optimise postures with the 3D tank as a constraint, and check the twist limits
4	Entry, storage and sensor head	Real head CAD and cable envelopes, sleeve contact, rear storage and retraction
5	Rear mechanism and integrated validation	Stage sizing, drive and locks, then a full integrated check

Sources and reproduction

All scripts, data and figures are in ASRC_kevin/telescopic_planar_study/backbone_20261108 (README sections 3 to 7). This report is generated by ASRC_kevin/telescopic_planar_study/reports/build_report_2026_11_09.py.